

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

PHY 411 — Past Questions & Materials
by Renance

Course: PHY 411
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 6
Levels: 200, 400

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Paper 1 of 6 — PHY 411 Statistical and Thermal physics

Level 400 • Second Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

COLLEGE OF SCIENCE

DEPARTMENT OF PHYSICS

SECOND SEMESTER EXAMINATION.

Course Code: PHY 411 Course Title: Statistical and Thermal Physics

Course Unit: 3 Units Session: 2023/2024

Time Allowed: 2 hrs 30 mins 29/4/2024

Instruction: Answer any four questions

1. (a) Define the following terms and write the mathematical relations for them (i) Boltzmann factor (ii) multiplicity ratio (iii) partition function (iv) average energy (3 marks each)

(b) Consider a single particle system with five states. There is one state with 4 energy and 2 states with energy 2ϵ . The system is in equilibrium with a heat bath at temperature T (a) calculate the partition function for the system (b) calculate the mean energy and heat capacity of the system as functions of temperature (c) what is the relative probability of the system having energies ϵ and 2ϵ (d) show that the partition function is given as $Z = 1 + 2e^{-2\beta\epsilon}$ © 5 marks} Y

2. (2) Define the Maxwell Boltzmann distribution. (2 marks) -
(6) Calculate the number of micro-states for eight non-interacting particles arranged into four bins such that the configuration of particles is $\{6,0,0,2\}$. (3 as marks) we
Define the following terms with the mathematical relations where required
(i) Configuration (ii) Weight of configuration (iii) Probability of a state (iv) Most probable configuration (4 marks)
Obtain the sharpness of the product for a two-spin system in thermal contact in a model spin system (2 marks)

3. (a) Define the following (i) Ensemble (ii) Canonical ensemble (iii) Grand canonical ensemble (75 marks)
(b) Explain the difference between the absolute temperature and the fundamental temperature. Write the symbols and the unit for each (4 marks) .
(c) Consider a classical system of $N \gg 1$ weakly interacting particles, each of which has energy given by $\epsilon = \frac{p^2}{2m} + Ax$ where x and p are the position and momentum of the particle. The system is in equilibrium at temperature T . (i) a

2 = 0, what is the molar specific heat of the system (ii) If A is not zero, but the quartic term is very small compared to the quadratic term, what will be the Beige

6%

variation in the specific heat of the system with the temperature. Use $ce'' = 1 + u$ for small u (6 marks)

4. (a) Define the Fermi-Dirac distribution function (3 marks)

(b) Using the Gibb's sum $77 = 14A \exp(-\beta \epsilon_i)$ obtain the mathematical relation for the * Fermi-Dirac distribution function. a5 marks)

(c) Six fermions are arranged in two compartments. The first compartment has seven cells, and the second compartment has eight cells of equal size. What is the total number of microstates for the macrostate (2,4) (6 marks)?

(d) Obtain the classical distribution function for an ideal gas in the classical limit. (4 marks)

5. (a) Define the Bose-Einstein distribution function (3 marks)

(b) State three properties of the Bose-Einstein distribution function (3 marks)

(c) Obtain the mathematical relation for the Bose-Einstein distribution function for a system of non-interacting bosons in thermal and diffusive contact with a reservoir a5 marks)

partments. The first compartment has

(d) Seven bosons are arranged in two compartments of equal size. What is the

eight cells, and the second compartment has nine cells. What is the total number of microstates for the macrostate (3,4) (7 marks)?

6. (a) Define the four thermodynamic potentials with relevant mathematical relations (10 marks)

(b) State three fundamental assumptions of thermal Physics (3 marks)

(c) Let a 10g specimen at a temperature of 350 K be placed in thermal contact with an identical specimen at a temperature of 290 K. Find the quantity of energy transferred when the two specimens are placed in contact and come to equilibrium at the final temperature T_f . The specific heat of metallic copper over the temperature range 15°C to 100°C is approximately 0.389 J/g °C

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Paper 2 of 6 — PHY 411 - 2022/2023 - First Semester - Statistical and Thermal Physics

Level 200 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES EFFURUN



DEPARTMENT OF PHYSICS

Course title: PHY 411
Course code: Statistical and Thermal Physics
Time allowed: 45 minutes
Unit: 3 units
Marks: 20 mks

1. In a monoatomic ideal gas, $PV = nRT$, $u = \frac{3}{2}nRT$, and $C_v = \frac{3}{2}nR$. Use these relations to derive an expression for u starting with entropy. (4 marks)
2. Air is compressed from an initial condition of 1 bar and 25°C to a final state of 5 bar and 25°C by three different mechanically reversible processes in a closed system:
- Heating at constant volume followed by cooling at constant pressure.
 - Isothermal compression.
 - Adiabatic compression followed by cooling at constant volume.

Assume air to be an ideal gas with the constant heat capacities, $C_v = (5/2)R$ and $C_p = (7/2)R$. Calculate the work required, heat transferred, and the changes in internal energy and enthalpy of the air for each process. (10 marks)

3. Describe the following distribution function statistics; (i) Maxwell-Boltzmann statistics (ii) Bose-Einstein statistics (iii) Fermi-Dirac statistics

4. Using the Bose-Einstein distribution of state $n(E) = g(E)F_{BE}$, where the Bose-Einstein factor $F_{BE} = \frac{1}{e^{\beta E} - 1}$ derive the radiation law given as $\gamma(\lambda, T) = \frac{2\pi c^2 h}{\lambda^5} \frac{1}{e^{hc/\lambda kT} - 1}$

$E = \frac{hc}{\lambda}$

$E = \frac{h^2}{2m\lambda^2}$

$E = \frac{p^2}{2m}$

$p = \frac{h}{\lambda}$

$p = \frac{h}{\lambda} \left(n_1^2 + n_2^2 + n_3^2 \right)^{1/2}$

$p = \frac{h}{\lambda} \left(n_1^2 + n_2^2 + n_3^2 \right)^{1/2}$

Energy $E = pc$

$p = \frac{E}{c}$

$E = \frac{hc}{\lambda} \left(n_1^2 + n_2^2 + n_3^2 \right)^{1/2}$

$E = \frac{hc}{\lambda} \sqrt{r} \quad r = \frac{E^2 c^2}{h^2 c^2} = \frac{E^2}{h^2 c^2}$

$N_v = \sum_{\lambda} \frac{1}{e^{\beta E} - 1}$

$N_v = \sum_{\lambda} \frac{1}{e^{\beta hc/\lambda} - 1}$

$N_v = \sum_{\lambda} \frac{1}{e^{\beta hc/\lambda} - 1}$

$\gamma(\lambda, T) = \frac{dN_v}{d\lambda d\lambda} = \frac{8\pi \lambda^{-5}}{c} \frac{1}{e^{\beta hc/\lambda} - 1}$

$N_v = \sum_{\lambda} \frac{1}{e^{\beta hc/\lambda} - 1}$

$F_{BE} = \frac{1}{e^{\beta E} - 1} = \frac{1}{e^{\beta hc/\lambda} - 1}$

$N_v = \frac{8\pi \lambda^{-5}}{c} \frac{1}{e^{\beta hc/\lambda} - 1}$

$U(E) \rightarrow$ returning the equation in terms of energy distribution. (E/s)

Paper 3 of 6 — PHY 411 - 2022/2023 - First Semester - Statistical and Thermal Physics

Level 200 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES EFFURUN



DEPARTMENT OF PHYSICS

Course title: PHY 411
Course code: Statistical and Thermal Physics
Time allowed: 45 minutes
Unit: 3 units
Marks: 20 mks

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2. Air is compressed from an initial condition of 1 bar and 25°C to a final state of 5 bar and 25°C by three different mechanically reversible processes in a closed system:
- Heating at constant volume followed by cooling at constant pressure.
 - Isothermal compression.
 - Adiabatic compression followed by cooling at constant volume.

Assume air to be an ideal gas with the constant heat capacities, $C_v = (5/2)R$ and $C_p = (7/2)R$. Calculate the work required, heat transferred, and the changes in internal energy and enthalpy of the air for each process. (10 marks)

3. Describe the following distribution function statistics; (i) Maxwell-Boltzmann statistics (ii) Bose-Einstein statistics (iii) Fermi-Dirac statistics

4. Using the Bose-Einstein distribution of state $n(E) = g(E)F_{BE}$, where the Bose-Einstein factor $F_{BE} = \frac{1}{e^{\beta E} - 1}$ derive the radiation law given as $\gamma(\lambda, T) = \frac{2\pi c^2 h}{\lambda^5} \frac{1}{e^{hc/\lambda kT} - 1}$

$E = \frac{hc}{\lambda}$

$E = \frac{h^2}{2m\lambda^2}$

$E = \frac{p^2}{2m}$

$p = \frac{h}{\lambda}$

$p = \frac{h}{\lambda} \left(n_1^2 + n_2^2 + n_3^2 \right)^{1/2}$

$p = \frac{h}{2L} \left(n_1^2 + n_2^2 + n_3^2 \right)^{1/2}$

Energy $E = pc$

$p = \frac{E}{c}$

$E = \frac{hc}{2L} \left(n_1^2 + n_2^2 + n_3^2 \right)^{1/2}$

$E = \frac{hc}{2L} \sqrt{r} \quad r = \frac{E^2 L^2}{h^2 c^2} = \frac{2L^2 E^2}{h^2 c^2}$

$N_v = \sum_{\lambda} \frac{1}{e^{\beta E} - 1}$

$N_v = \frac{8\pi V}{h^3 c^3} \int_0^\infty \frac{E^2 dE}{e^{\beta E} - 1}$

$N_v = \frac{8\pi V}{h^3 c^3} \int_0^\infty \frac{E^2 dE}{e^{E/kT} - 1}$

$\gamma(E) = \frac{dN_v}{dE} = \frac{8\pi V E^2}{h^3 c^3} \frac{1}{e^{E/kT} - 1}$

$N_v = \sum_{\lambda} \frac{1}{e^{\beta E} - 1}$

$N_v = \frac{8\pi V}{h^3 c^3} \int_0^\infty \frac{E^2 dE}{e^{E/kT} - 1}$

$F_{BE} = \frac{1}{e^{\beta E} - 1} = \frac{1}{e^{E/kT} - 1}$

$N_v = \frac{8\pi V E^2}{h^3 c^3} \frac{1}{e^{E/kT} - 1}$

$E = \frac{hc}{\lambda}$

$U(E) \rightarrow$ returning the equation in terms of energy distribution. (E/s)

Paper 4 of 6 — PHY 411 - 2022/2023 - First Semester - Statistical and Thermal Physics

Level 200 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES EFFURUN



DEPARTMENT OF PHYSICS

Course title: PHY 411
Course code: Statistical and Thermal Physics
Time allowed: 45 minutes
Unit: 3 units
Marks: 20 mks

1. In a monoatomic ideal gas, $PV = nRT$, $u = \frac{3}{2}nRT$, and $C_v = \frac{3}{2}nR$. Use these relations to derive an expression for u starting with entropy. (4 marks)
2. Air is compressed from an initial condition of 1 bar and 25°C to a final state of 5 bar and 25°C by three different mechanically reversible processes in a closed system:
- Heating at constant volume followed by cooling at constant pressure.
 - Isothermal compression.
 - Adiabatic compression followed by cooling at constant volume.

Assume air to be an ideal gas with the constant heat capacities, $C_v = (5/2)R$ and $C_p = (7/2)R$. Calculate the work required, heat transferred, and the changes in internal energy and enthalpy of the air for each process. (10 marks)

3. Describe the following distribution function statistics; (i) Maxwell-Boltzmann statistics (ii) Bose-Einstein statistics (iii) Fermi-Dirac statistics

4. Using the Bose-Einstein distribution of state $n(E) = g(E)F_{BE}$, where the Bose-Einstein factor $F_{BE} = \frac{1}{e^{\beta E} - 1}$ derive the radiation law given as $\gamma(\lambda, T) = \frac{2\pi c^2 h}{\lambda^5} \frac{1}{e^{hc/\lambda kT} - 1}$

$E = \frac{hc}{\lambda}$

$E = \frac{h^2}{2m\lambda^2}$

$E = \frac{p^2}{2m}$

$p = \frac{h}{\lambda}$

$p = \frac{h}{2\lambda}$

$p = \frac{h}{\lambda}$

Energy $E = pc$

$p = \frac{E}{c}$

$E = \frac{hc}{\lambda}$

$E = \frac{hc}{\lambda}$

$N_v = \frac{8\pi V}{h^3} \int_0^\infty \frac{E^2 dE}{e^{\beta E} - 1}$

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$\gamma(E) = \frac{dN_v}{dE} = \frac{8\pi V}{h^3} \frac{E^2}{e^{\beta E} - 1}$

$\gamma(E) = \frac{dN_v}{dE} = \frac{8\pi V}{h^3} \frac{E^2}{e^{\beta E} - 1}$

$F_{BE} = \frac{1}{e^{\beta E} - 1}$

$N_v = \frac{8\pi V}{h^3} \int_0^\infty \frac{E^2 dE}{e^{\beta E} - 1}$

$U(E) \rightarrow$ returning the equation in terms of energy distribution. (E/s)

Paper 5 of 6 — PHY 411 TEST PQ COMPUTATIONAL PHYSICS

Level 400 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES EFFURUN



DEPARTMENT OF PHYSICS

Course title: PHY 411
Course code: Statistical and Thermal Physics
Time allowed: 45 minutes
Unit: 3 units
Marks: 20 mks

1. In a monoatomic ideal gas, $PV = nRT$, $u = \frac{3}{2}nRT$, and $C_v = \frac{3}{2}nR$. Use these relations to derive an expression for u starting with entropy. (4 marks)
2. Air is compressed from an initial condition of 1 bar and 25°C to a final state of 5 bar and 25°C by three different mechanically reversible processes in a closed system:
- Heating at constant volume followed by cooling at constant pressure.
 - Isothermal compression.
 - Adiabatic compression followed by cooling at constant volume.

Assume air to be an ideal gas with the constant heat capacities, $C_v = (5/2)R$ and $C_p = (7/2)R$. Calculate the work required, heat transferred, and the changes in internal energy and enthalpy of the air for each process. (10 marks)

3. Describe the following distribution function statistics; (i) Maxwell-Boltzmann statistics (ii) Bose-Einstein statistics (iii) Fermi-Dirac statistics

4. Using the Bose-Einstein distribution of state $n(E) = g(E)F_{BE}$, where the Bose-Einstein factor $F_{BE} = \frac{1}{\beta_{BE} \exp(\beta E) - 1}$ derive the radiation law given as $\gamma(\lambda, T) = \frac{2\pi c^2 h}{\lambda^5} \cdot \frac{1}{e^{hc/\lambda kT} - 1}$

$$E = \frac{h^2}{8mL^2} (n_1^2 + n_2^2 + n_3^2)$$

$$E = \frac{p^2}{2m}$$

$$p^2 = \frac{h^2}{4mL^2} (n_1^2 + n_2^2 + n_3^2)$$

$$p = \frac{h}{2L} (n_1^2 + n_2^2 + n_3^2)^{1/2}$$

$$\text{Energy } E = pc$$

$$p = \frac{E}{c}$$

$$E = \frac{hc}{2L} (n_1^2 + n_2^2 + n_3^2)^{1/2}$$

$$E = \frac{hc}{2L} \cdot r \quad r = \frac{E^2 c}{hc} = \frac{E^2}{hc}$$

$$u(E) = \frac{8\pi E^2}{h^3 c^3} \frac{1}{e^{E/kT} - 1}$$

$$u(E) dE = \frac{8\pi E^2}{h^3 c^3} \frac{dE}{e^{E/kT} - 1}$$

$$E = \frac{hc}{\lambda}$$

$$N_v = \frac{1}{\lambda^3} \left(\frac{4\pi r^3}{3} \right)$$

$$N_r = \frac{8\pi r^3}{3 \lambda^3 c^3} = \frac{8\pi E^3}{3 h^3 c^3}$$

$$N(E) = \frac{dN_r}{dE} \cdot \text{for } h^3 c^3$$

$$g(E) = \frac{dN_r}{dE} = \frac{8\pi E^2}{h^3 c^3} \cdot \frac{1}{\lambda^3 c^3}$$

$$N(E) = g(E) \cdot F_{BE} \quad B = \frac{1}{kT}$$

$$F_{BE} = \frac{1}{e^{BE} - 1} = \frac{1}{e^{E/kT} - 1}$$

$$N(E) = \frac{8\pi E^2}{h^3 c^3} \frac{1}{e^{E/kT} - 1}$$

$$E = \frac{hc}{\lambda}$$

$$u(E) \rightarrow \text{rearrange the equation in term of Energy distribution } (E/s)$$

Paper 6 of 6 — statistical and thermal physics PHY 411 PQ

Level 400 • First Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

COLLEGE OF SCIENCE

DEPARTMENT OF PHYSICS

SECOND SEMESTER EXAMINATION.

Course Code: PHY 411 Course Title: Statistical and Thermal Physics

Course Unit: 3 Units -- Session: 2023/2024

Time Allowed: 2 hrs 30 mins 29/4/2024

Instruction: Answer any four questions

1. (a) Define the following terms and write the mathematical relations for them (i)

Boltzmann factor (ii) multiplicity ratio (iii) partition function (iv) average energy
(3 marks each)

(b) Consider a single particle system with five states. There is one state with energy ϵ and 2 states with energy 2ϵ . The system is in equilibrium with a heat bath at temperature T (a) calculate the partition function for the system (5)

calculate the mean energy and heat capacity of the system as functions of temperature (c) what is the relative probability of the system having energies 2ϵ and ϵ (d) show that the partition function is given as

$$Z = \sum_i e^{-\beta \epsilon_i}$$

$$D = \frac{1}{N!} \left(\frac{2\pi m k_B T}{h^2} \right)^{3/2} V$$

2. (a) Define the Maxwell Boltzmann distribution. (2 marks)

(b) Calculate the number of micro-states for eight non-interacting particles as arranged into four bins
(3 marks)

Define the following terms with the mathematical relations where required

(i) Configuration (ii) Weight of configuration (iii) Probability of a state (iv)

Most probable configuration (4 marks)

in the sharpness of the distribution

such that the configuration of particles is $\{600, 2\}$.

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ical

i) Grand canonical

tet

variation in the Specific heat of the system with the temperature for small y (6 marks)

4. (a) Define the Fermi-Dirac distribution

istribution function

(b) Using the Gibb's sum 7

(3 marks)

=! +2exp(- ve) obtain the mathematical

Fermi-Dirac distribution function, as

(c)

Six fermions are arranged in two comp

Seven cells, and the Second compartm

the total number of microstates for t

(d) Obtain the classical distribution func

(4 marks)

he macrostate (2,4) (6

tion for an ideal gas in

5. (a) Define the Bose-Einstein distribution function (3 marks)

(6) State three Properties of the Bose-Einstein distribution funct,

(c) Obtain the mathematical relation i

modynamic potentials with relevant mathematica

(10 marks)

b) State three fundamen

c) Letaio 9 Specimen at

with an identical specimen at a temperc